COMPUTER SCIENCE & DA

Data Structure through Python

QUEUES AND HASH TABLES

Discussion Notes

DPP- 01



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[MCQ]

#Q In queue, Deletion is happens at the _____end.

A Rear

B Front

C Either rear or front

D More than one of the above

E None of the above



[MCQ]

#Q Carefully observe the below given queue

REAR	22	57	~~96~~	~~68~~	~~19~~	~~15~~	~~28~~	FRONT

How many dequeue operations needed to delete the largest element from the queue?

28, 15, 19, 68, 96

A 1

B 3

C 5

D More than one of the above

E None of the above

[MCQ]

#Q A queue is implemented using an array such that ENQUEUE and DEQUEUE operations are performed efficiently. Which one of the following statements is CORRECT (n refers to the number of items in the queue)?

A Both operations can be performed in 0(1) time

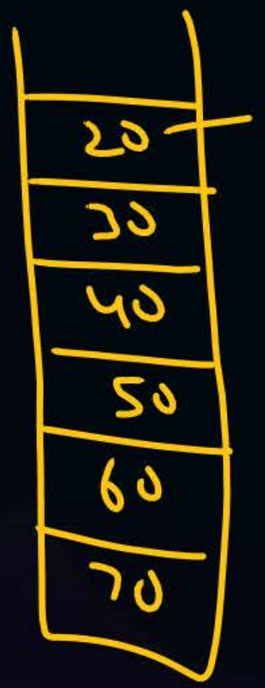
B At most one operation can be performed in 0(1) time but the worst case time for the other operation will be Ω(n)

C The worst-case time complexity for both operations will be Ω(n)

D Worst case time complexity for both operations will be Ω(log n)

[MCQ]

#Q A priority queue Q is used to implement a stack that stores characters. PUSH (C) is implemented INSERT (Q, C, K) where K is an appropriate integer key chosen by the implementation. POP is implemented as DELETEMIN(Q). For a sequence of operations, the keys chosen are in



A Non-increasing order

B Non-decreasing order

C Strictly increasing order

D Strictly decreasing order

[MCQ]

#Q The best data structure to check whether an arithmetic expression has balanced parentheses is a



A Queue

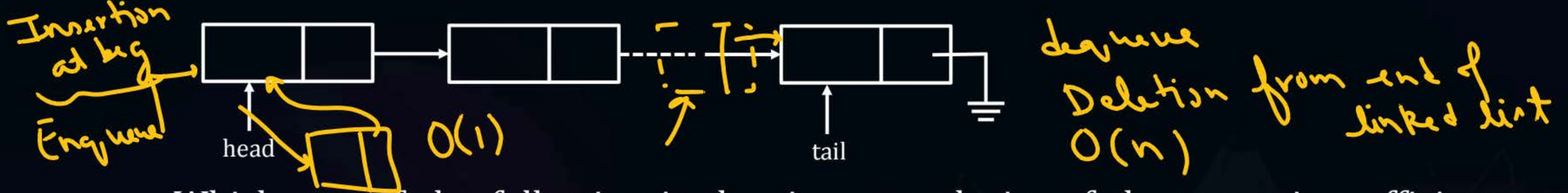
B Stack

C Tree

D List

[MCQ]

#Q A queue is implemented using a non-circular singly linked list. The queue has a head pointer and a tail pointer, as shown in the figure. Let n denote the number of nodes in the queue. Let enqueue be implemented by inserting a new node at the head, and dequeue be implemented by deletion of a node from the tail.



Which one of the following is the time complexity of the most time-efficient implementation of enqueue and dequeue, respectively, for this data structure?

A $\Theta(1), \Theta(1)$

B $\Theta(1), \Theta(n)$

C $\Theta(n), \Theta(1)$

D $\Theta(n), \Theta(n)$

THANK - YOU